CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2012

EE309 Electrical Power Utilization and Traction

Date: 10.05.2012, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 What is Load Equalization? And Derive the formula $T_m = T_L - (T_L - T_0)e^{\frac{-t}{k}}$ from first Principles. **[07]**

Q - 2 (a) Explain Regenerative Braking Of 3-phase Induction Motor. **[05]**

(b) Explain the Methods of Speed Control of DC Shunt Motor. **[05]**

(c) A 6 pole lap wound shunt motor has 500 conductors in the armature. The resistance of armature path is 0.05Ω. The resistance of shunt field is 25 Ω. Find the speed of the motor when it takes 120A from DC mains of 100V supply. Flux per pole is 2×10^{-2}wb. **[04]**

OR

Q - 2 (a) State the factor Affecting the selection of Electric motor. **[05]**

(b) Explain the Four Quadrant Operation of Electric motors. **[05]**

(c) A 4 Pole, 50 HZ Induction motor has a Flywheel on its Shaft. Total Inertia at the motor shaft is 1000 kg-m^3. Load torque is 100 kg-m for 10 seconds followed by a no-load period long enough for the flywheel to regain its full speed. Motor has a slip of 6% at a torque of 50 kg-m. Calculate the speed at the end of deceleration period. Assume motor speed-torque characteristic to be a straight line in the region of interest and neglect friction and windage losses. **[04]**

Q - 3 (a) Give the classification of various electric heating methods along with brief account of their working principles. **[05]**

(b) Write in brief about the vertical core type induction furnace. **[05]**

(c) Explain the application of high frequency eddy current heating. **[04]**

OR

Q - 3 (a) Discuss the following application of dielectric heating [05]

1) Heating of raw plastics.

2) Gluing of wood.

3) Food Processing.

(b) Write the types of Arc Furnaces. And explain any one of them. [05]

(c) State the first law of faraday, for Electro Deposition. [04]

SECTION – II

Q - 4 Explain solid angle. Also state and Explain the laws of Illumination. [07]

Q - 5 (a) Explain the working of Fluorescent tube with the help of circuit Diagram giving the function of various parts. [07]

(b) Write a short note on Street Lighting. [07]

OR

Q - 5 (a) Draw a simplified speed time curves and Derive the Equation for Quadrilateral speed-time curve. [07]

(b) The speed time curve of a train consists of [07]

1) Uniform acceleration of 6 km/h/s for 25 s.

2) Free running for 10 minutes.

3) Uniform Deceleration of 6 km/h/s to stop the train.

4) A stop of 5 minutes.

Find the distance between the stations, The average and schedule speeds.

Q - 6 (a) What is Crest speed; Average speed & Schedule speed and also Explain the Factors affecting Schedule speed. [07]

(b) Explain in brief about 'Tractive effort' for overcoming the effect of gravity and Tractive effort for overcoming train Resistance. [07]

OR

Q - 6 (a) Explain the terms 'Dead weight', 'Effective weight', and 'Adhesive weight' in a locomotive. [07]

(b) Give a comparison between AC & DC Welding. [07]

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